WildTrace: Photo-to-Pen Robot Drawing with Learned Path-Following Policies and a Calibrated Digital Twin

Author One, Author Two, Author Three
Affiliation
City, Country
{author}@example.edu

***Abstract*—We present WildTrace, an end-to-end system that turns photographs of animals and faces into pen-stroke trajectories and draws them with a 7-DoF UFACTORY xArm 7, both in a calibrated NVIDIA Isaac Sim twin and on the physical arm. A medallion data pipeline uses a diffusion model to redraw each photo as a clean outline, a vision-language judge with OpenCV prescreening to reject bad outlines, and a vectoriser that emits a few ordered strokes, yielding 2,874 validated trajectories over six animal classes. On top of this data we train an Action Chunking Transformer (ACT) that maps joint state and a short window of the upcoming path to joint steps, using only kinematic inverse-kinematics labels with noise injection. On a held-out set of 48 drawings in the twin at 1 g, ACT finishes 48/48 drawings against 45/48 for a Lula IK baseline, lowers mean path RMSE from 0.277 mm to 0.192 mm (lower on all 45 paired drawings, Wilcoxon $p < 10^{-12}$), halves joint jerk, and is unaffected by $\pm$12 mm paper height and 21 mm planar placement errors. A streaming IK baseline without the settle gate gains only 0.010 mm, so the improvement comes from the learned policy. A fine-tuned SmolVLA re-planned every tick is slightly more accurate still (0.180 mm) but jerkier and twice as slow to query, and a rule-based supervisor that hands stalls and ink gaps to IK repairs injected ink skips (87 redraws for 105 skips, none abandoned) at under 0.011 mm cost. On the physical arm, ACT completes 30/30 animal drawings at 0.532 mm RMSE and three face portraits at 0.497 mm, about 2.7$\times$ the twin, a gap that we analyse.**

***Index Terms*—robot drawing, imitation learning, action chunking, vision-language-action models, digital twin, sim-to-real**

I. Introduction

Robot drawing is a compact testbed for precise contact-rich path following: the pen must stay on a sheet a fraction of a millimetre away, follow tight curves without overshoot, and lift and land cleanly between strokes. Every error is visible on paper. Classic pipelines solve inverse kinematics (IK) per waypoint and wait for the arm to settle [1], [2], which is reliable but slow and leaves a visible tremor. Learned visuomotor policies such as ACT [3], diffusion policy [4] and vision-language-action (VLA) models such as SmolVLA [5] promise smoother motion, but are usually evaluated by task success rather than by sub-millimetre path accuracy on open-ended content.

WildTrace closes the loop from an arbitrary photograph to ink on paper (Fig. 1) and uses it to compare controllers on the same drawings under identical physics. Our contributions are:

1) A photo-to-trajectory pipeline with a generate-judge-retry loop, which we use to build 2,874 validated drawing trajectories and a 48-drawing held-out test set (Sec. III).
2) A path-conditioned ACT policy trained only on kinematic IK labels with noise injection and a capped progress projection, which beats IK in a calibrated 1 g twin on accuracy, completion and smoothness, is insensitive to paper placement, and keeps its lead over a streaming IK baseline that shares its progress rule and pace (Sec. IV, V).
3) A like-for-like comparison with a fine-tuned SmolVLA on the same data: re-planned every tick it matches and slightly exceeds ACT on the test set, at higher jerk and latency, and a negative result on camera-conditioned gap repair.
4) A hybrid supervisor that repairs skipped ink using the run's own bookkeeping, and a physical-arm evaluation on 30 animal drawings and three portraits that quantifies the sim-to-real gap.

II. Related Work

Robot drawing and painting. Portrait-drawing robots date back to humanoid systems that extract contours and replay them with IK [1] and to Paul, which emulates an artist's sketching process [2]. Recent systems differ mainly in what they draw and how fast. RoboCoDraw turns a face into a GAN avatar and orders its strokes with a genetic algorithm, drawing in 43 s on a UR5 [9]; AiSketcher draws synthesised portraits with a 3-DoF arm in about two minutes [10]; Nasrat et al. draw portraits in 80 s with line width set by pen height [11]; Piktor-O-Bot adds force-controlled shading on a UR5 and shapes speed by path curvature [12], [13]; TSP-Bot draws single-path stipple art with TRAC-IK [14]; and line art can be parsed into stroke graphs for a desktop arm [15]. These systems execute planned paths and report drawing time, style or viewer preference rather than tracking accuracy. Learned stroke models [16], [17] plan in image space, and FRIDA [18] and Spline-FRIDA [19] learn stroke appearance to shrink the sim-to-real gap of the painted image. Closest to our control question, Lee et al. learn motor control with hierarchical reinforcement learning instead of IK, but need pen-tip feedback to



Fig. 1. WildTrace data pipeline on one Open Images [6] photo. Bronze: fetch the image, mask and license, then curate and label (BioCLIP [7]). Silver: isolate the subject and redraw it as an outline with FLUX [8], judged and retried. Gold: the accepted outline becomes a few ordered pen strokes on a normalised canvas (numbered markers show stroke starts).

correct the learned controller on a real UR5e [20]. We take the strokes as given and measure how precisely learned policies follow them against IK, at sub-millimetre resolution.

Imitation learning for fine control. ACT [3] predicts chunks of future actions with a transformer; diffusion policy [4] models multimodal action distributions. DART [21] injects noise during demonstration to cover off-path states and DAgger [22] aggregates on-policy corrections. We use DART-style noise with a synthetic IK teacher, and DAgger rounds for camera policies.

Vision-language-action models. SmolVLA [5] reuses a pretrained vision-language backbone for control, with a flow-matching action head [23]. We fine-tune SmolVLA on the same data as ACT to test whether a pretrained generalist helps a precision task.

Sim-to-real. Domain randomisation [24] widens the training distribution to cover the real system; we instead calibrate the twin to the lab arm and randomise only paper placement.

III. Data Pipeline: Photo to Pen Strokes

The pipeline is a local-first medallion design with one script per stage and append-only NDJSON manifests, so a stage reruns only for rows whose configuration or input hash changed.

Bronze. Images, segmentation masks and licenses are fetched from Open Images V7 for six classes (bird, cat, dog, fish, frog, horse), de-duplicated, filtered for quality and viewing angle, and labelled with BioCLIP.

Silver. The subject is cropped by its mask and redrawn as a black-on-white outline by FLUX.1-schnell. A LangGraph [25] loop validates each outline with an OpenCV prescreen (foreground ratio, component count, contour size, fragmentation) followed by a VLM judge, and retries with a new seed on failure.

Gold. The best outline per pose is skeletonised and vectorised into a few ordered strokes, resampled and normalised to a unit canvas. The result is 2,874 validated trajectories. A stratified 48-drawing gold test set (9 bird, 12 cat, 13 dog, 4 fish, 5 frog, 5 horse; mean 4.6 strokes, range 2 to 8, mean 370 points) is never used for training. Faces follow a separate CPU path: MediaPipe [26] face landmarks and hair/body segmentation are traced into a pen-sketch portrait.

TABLE I
Outline generator and judge study (20 subjects, referee Qwen3-VL-235B).

Generator	Referee pass	BioCLIP top-1
FLUX.1-schnell	**20/20**	**75%**
OmniGen2	6/20	30%
Judge	**Agreement**	**Cohen's κ**
Qwen2.5-VL-7B (local)	passes 9/9 blanks	0.00
Qwen3-VL-8B	**92.5%**	**0.83**

Fig. 2 shows five portraits from this path next to their source photographs.

Choosing the generator and the judge. Table I summarises a 20-subject study against a strong referee (Qwen3-VL-235B [27]). FLUX produced acceptable outlines for all 20 subjects against 6/20 for OmniGen2 [28] (paired sign test $p = 10^{-4}$), whose failures were mostly blank pages. The local Qwen2.5-VL-7B [29] judge originally in production agreed with the referee no better than chance and passed every blank page; we therefore switched to Qwen3-VL-8B, which agrees with the referee on 92.5% of outlines.

Why open-weight models, not frontier APIs. Every generator and judge in the pipeline is an open-weight model that runs on the lab's single 16 GB laptop GPU, and we did not evaluate proprietary frontier models in either role. Four reasons drove this. (i) *Volume and cost*: the dataset took thousands of generate-judge calls with up to three attempts per subject, at zero marginal cost locally; a hosted open 8B judge costs about \$0.00004 per call, which keeps even hosted runs negligible. (ii) *Privacy*: WildTrace Studio turns uploaded photographs of people into portraits, and a local pipeline never sends those faces to a third party. (iii) *Reproducibility*: pinned open weights give the same verdicts on a rerun, whereas a hosted frontier model can change under the same name. (iv) *Sufficiency*: an 8B open judge already agrees with a 235B open referee at $\kappa = 0.83$, so a larger judge has little room to improve the accept/reject decision. The cost is that we cannot report how a frontier judge would score; the referee is itself an open model.



Fig. 2. Face portraits: Open Images source photographs (top) and the pen-sketch line art traced from MediaPipe landmarks and segmentation (bottom), 15 to 19 strokes each. Photographs CC BY 2.0 via Flickr, by Byron Gonzalez, MangakaMaiden Photography, Luca Boldrini, JJP Bolger Photography and Walt Stoneburner (left to right).



(a) Lab-arm twin (b) Physical xArm 7 (c) Real portrait

Fig. 3. (a) Isaac Sim twin built from the lab arm's calibrated URDF, gripper and measured paper placement, at 1 g. (b) The physical xArm 7 drawing a WildTrace bird. (c) A face portrait drawn by ACT on the physical arm with a marker (run 7 in Table VIII).

IV. Robot, Twin and Controllers

Hardware and twin. A UFACTORY xArm 7 holds a pen in its parallel gripper over a US Letter sheet. The Isaac Sim [30] twin (Fig. 3a) uses the arm's calibrated URDF, the measured paper pose and pen touch height, Earth gravity, and a stiff PhysX position servo (stiffness 7.5×10^5, damping 2050) standing in for the vendor joint controller. Two RealSense D435i cameras (wrist and overhead) were located by depth plane fitting and planar PnP on the sheet corners, whose heights above the paper agree to within 1.6%. Each drawing runs a fixed phase machine: approach, lower, draw, lift for every stroke, with short pen-up hops between nearby strokes.

Metrics. Let $\{p_i\}_{i=1}^{N}$ be the planned pen-down samples and $\{e_j\}$ the executed pen-down pen-tip positions, both in 3-D. The nearest-path error of a planned sample is

$$d_i = \min_j \|p_i - e_j\|_2, \tag{1}$$

and we report the path RMSE $\left(\frac{1}{N}\sum_i d_i^2\right)^{1/2}$, the 95th percentile and the maximum of d_i. Because (1) only penalises missing a planned point, we also report the reverse direction, from each executed sample to the plan, which catches stray ink. Joint jerk is the RMS over pen-down samples of $\|\Delta^3 q\|/\Delta t^3$, the third difference of joint positions within a stroke. Corner RMSE is the tracking error at waypoints flagged as corners, and heading error compares the direction of plan and pen over a 4 mm chord.

IK baseline. NVIDIA Lula IK solves joint targets for each waypoint of the resampled path (at most 5 mm apart, densified at corners); joint commands are slew-limited, and the run advances only when the measured tip is within tolerance and its velocity has settled. A waypoint timeout fails the drawing safely. This is the controller the system shipped with and the teacher that labels the policy data.

Streaming IK baseline. To separate learning from the settle gate, a second IK baseline uses exactly ACT's progress rule (the first waypoint ahead of the pen's projection, Sec. IV) with no settle gate, and solves Lula IK every tick for a carrot a fixed arc length L ahead of the pen (pure pursuit). The carrot advances along the path no faster than ACT's median pen speed, 12.4 mm s^{-1}, so both controllers draw at the same pace; we also run it at plain IK's median pen speed, 6.9 mm s^{-1}. On six drawings outside the test set, L between 0.5 and 2 mm made no difference because the speed limit binds, so we use $L = 0.5$ mm. Without the speed limit, pure pursuit raced through a drawing in a sixth of the time at 0.69 mm RMSE.

ACT path follower. At control tick t the policy observes the robot state $s_t = [\, q_t,\ \dot{q}_t,\ u_{t-1}\,] \in \mathbb{R}^{21}$ (joint positions, velocities and previous joint command) and a path context

$$c_t = \left[\, x_t,\ g_t - x_t,\ \{w_{t,k} - x_t\}_{k=1}^{10},\ \pi_t \,\right] \in \mathbb{R}^{37}, \tag{2}$$

where x_t is the pen tip, g_t the current target, $w_{t,k}$ ten points spaced at equal arc length along the next 2 cm of path, and π_t the pen state. It predicts a chunk of $H = 10$ joint

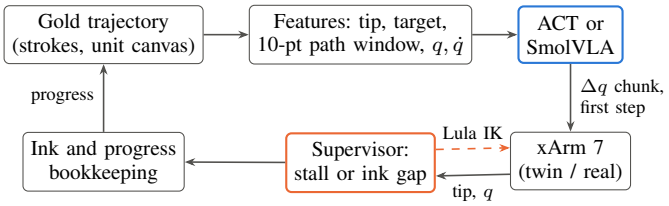


Fig. 4. Control loop. The policy runs every 60 Hz tick on a path window around the current progress; the supervisor hands control to Lula IK only on a stall or an uninked gap (dashed), then hands back.

increments $\hat{\delta}_{t:t+H-1}$, trained with an L1 loss against the teacher's increments, and only the first is executed:

$$u_t = u_{t-1} + \alpha\,\hat{\delta}_t, \qquad \alpha = 0.5. \tag{3}$$

Absolute joint targets were mm-imprecise even when overfitting one drawing, so the policy predicts increments. There is no image input. We use ACT [3] via LeRobot [31] with $d = 256$, 4 encoder and 1 decoder layers, no CVAE, no dropout, learning rate 3×10^{-4}, batch 256 and 40k steps. On a single-drawing overfit gate these choices cut first-action error from 33-45% to 11% of the label, a 10-step chunk fit 2.5× better at p95 than a 30-step chunk, and executing only the first action of each chunk with re-planning every tick tracked better than temporal ensembling. At run time the policy is queried at 60 Hz and half of each predicted step is applied.

Labels without physics. Labels come from Lula IK walked through the same phase machine with no physics simulation (about 20k solves/s), three episodes per training drawing. Two of the three add DART-style command noise [21] and a measured-versus-commanded joint offset, so the policy sees sagged, off-path states with the corrective label. Path progress is the projection of the tip onto the path, capped at 1.5 mm ahead; without the cap progress jumped across narrow hairpins (8.1 mm worst-point error on one frog toe) and spike tips were rounded. The lab-arm model is trained with ±15 mm paper-height jitter. Fine-tuning on 1 g Isaac IK rollouts made tracking worse (0.74 to 0.91 mm on four drawings), because those labels carry IK's own sag.

SmolVLA. We fine-tune SmolVLA [5] on the same dataset (19.8M frames), feeding the full 58-value state through a direct-state MLP, 8 VLM layers, 256 px rendered path images and 20-action chunks (51M of 323M parameters trained). It is served with zero initial noise (better than averaging samples offline); with 3 flow-matching steps, a captured CUDA graph cuts sampling from about 225 ms to 30 ms per chunk.

Hybrid supervisor. A rule-based supervisor (Algorithm 1, Fig. 4) watches the run's own ink bookkeeping rather than the policy. If the drawing target has not advanced for 1.5 s, IK drives the next 5 mm and hands back. If at least 1.5 mm of path behind the pen has no ink within 1 mm (the labeller's coverage rule), IK goes back past the gap and forward again while the run's progress is held, then the policy resumes. Both rules need no camera, so they also work on the physical arm.

Algorithm 1 Hybrid supervisor, run every control tick

Require: policy π, path progress s, inked set $\mathcal{I}$, last advance time t_a

if $t - t_a > 1.5$ s **then**
  drive the next 5 mm of path with Lula IK; $t_a \leftarrow t$ {stall}
else if a stretch $[s_0, s_1] \subset [0, s]$ with $s_1 - s_0 \geq 1.5$ mm has no ink within 1 mm **then**
  hold s; IK goes back to s_0, retraces to s {gap}
  add the retraced path to $\mathcal{I}$; give up after a timeout
else
  $u_t \leftarrow u_{t-1} + \alpha\,\pi(s_t, c_t)_1$ {policy, Eq. (3)}
end if
update s by projecting the tip onto the path (capped at 1.5 mm ahead)

TABLE II
GOLD TEST SET, 48 HELD-OUT DRAWINGS, LAB-ARM TWIN AT 1 G, SAME CODE AND RUNNER FOR ALL FOUR CONTROLLERS. MEANS OVER FINISHED DRAWINGS; 95% BOOTSTRAP INTERVALS OVER DRAWINGS IN BRACKETS. S-IK IS THE STREAMING IK BASELINE AT ACT'S PACE.

Metric	Lula IK	S-IK	ACT	SmolVLA
Finished	45/48	**48/48**	**48/48**	**48/48**
Path RMSE (mm)	0.277	0.267	0.192	**0.180**
95% CI	.271-.283	.262-.272	.190-.194	.176-.184
Path p95 (mm)	0.398	0.373	0.335	**0.327**
Path max (mm)	0.778	0.953	**0.776**	0.871
Reverse RMSE (mm)	0.397	0.401	0.374	**0.356**
Corner RMSE (mm)	0.366	0.608	**0.325**	0.330
Heading, median (°)	0.51	**0.41**	1.36	1.10
Jerk RMS (rad s^{-3})	757	745	**390**	521
Samples settling (%)	51	48	18	**17**
Median speed (mm s^{-1})	6.9	7.8	12.4	**14.1**
Sim. draw time (s)	76.8	101.0	73.4	**71.3**
Latency p50 (ms)	-	-	**9.1**	17.7

V. RESULTS

All simulated runs use the lab-arm twin at 1 g. The main comparison (Table II, Fig. 5) runs all four controllers on all 48 gold drawings with the same code, scene and batch runner; per-run metrics and plots are kept with the code.

ACT against IK. ACT finishes every drawing; IK fails three on waypoint timeouts. On the 45 drawings both finish, ACT has lower path RMSE on all 45: the paired reduction is 0.085 mm (95% CI 0.079 to 0.091), or 31%, with two-sided Wilcoxon signed-rank $p = 5.7 \times 10^{-14}$. The advantage holds in every class (Table III). It is not bought with stray ink: the reverse, pen-to-plan error is also lower (0.374 against 0.397 mm). Over pooled samples (Fig. 5b) the median error falls from 0.256 to 0.141 mm. ACT halves joint jerk (Fig. 5c) and removes the per-waypoint tremor visible in Fig. 6. IK keeps a lower heading error (median 0.51° against 1.36°), since it settles on each waypoint: its pen is settling on 51% of pen-down samples against 18% for ACT, and moves at a median 6.9 against 12.4 mm s^{-1}. Total draw time is similar (77 against 73 s simulated) because the phase machine's approach, lower and lift moves are shared. The policy server answers in 9.1 ms median, inside the 16.7 ms tick. Retraining

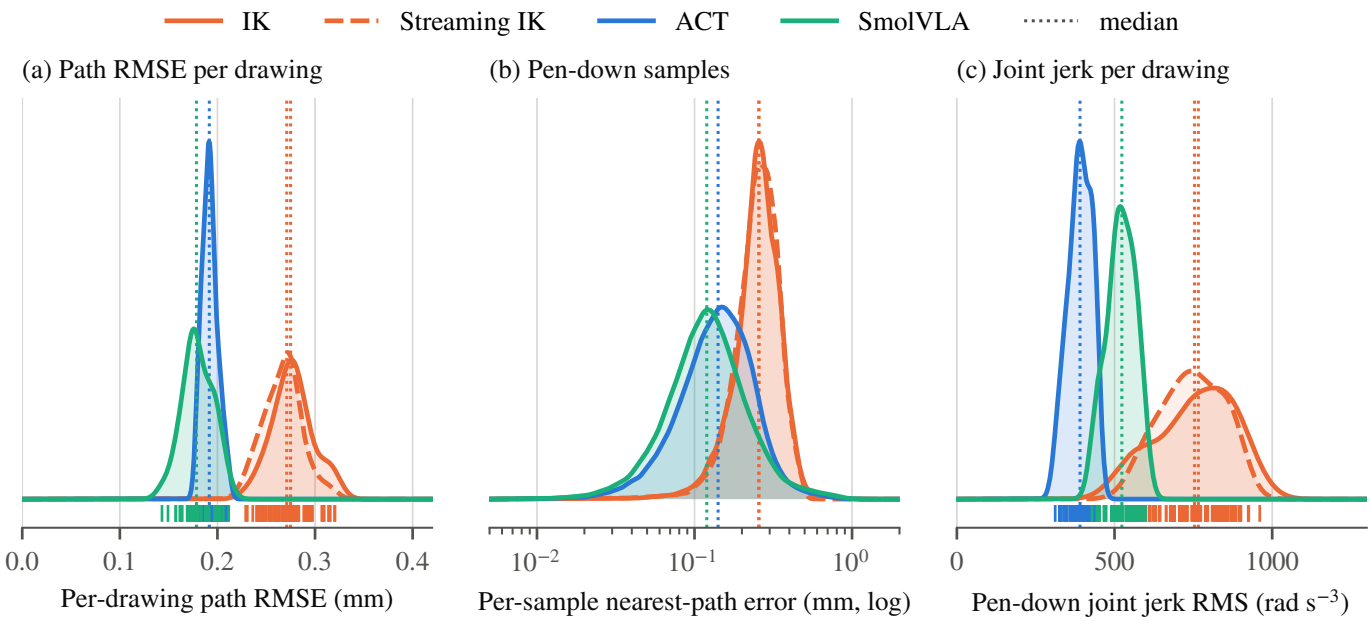


Fig. 5. Kernel density estimates (Gaussian kernel, Scott bandwidth) over the gold set for the four controllers; dotted lines are medians and ticks are individual drawings (n = 45 / 48 / 48 / 48 drawings, 71k / 76k / 76k / 76k pen-down samples for IK / streaming IK / ACT / SmolVLA). (a) Per-drawing path RMSE: the two learned policies separate cleanly from both IK variants. (b) Nearest-path error of every planned pen-down sample (densified plan): medians 0.256, 0.257, 0.141 and 0.120 mm. (c) Pen-down joint jerk: ACT is smoothest, SmolVLA between ACT and IK.

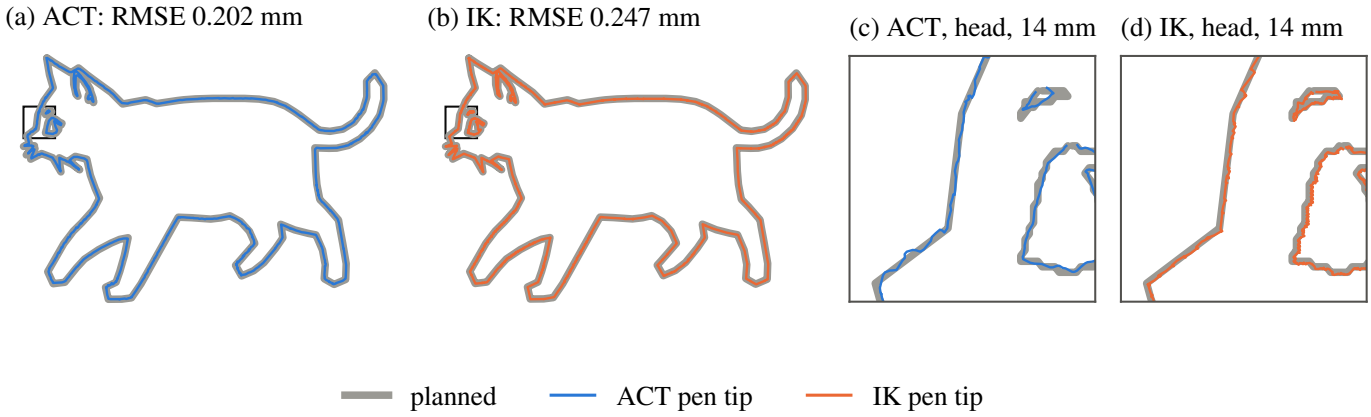


Fig. 6. Planned path (grey) against the executed pen tip for the same gold cat, ACT (blue) and IK (orange), top-down. (c, d) A fixed 14 mm window on the head: IK's stop-and-settle at each waypoint leaves a tremor of small hooks along the stroke; ACT stays smooth.

TABLE III
GOLD TEST SET PER CLASS: MEAN PATH RMSE IN MM (FINISHED / TOTAL WHERE NOT ALL FINISHED).

Class	n	Lula IK	S-IK	ACT	SmolVLA
Bird	9	0.266	0.260	0.191	**0.184**
Cat	12	0.275	0.267	0.195	**0.188**
Dog	13	0.282 (12)	0.270	0.189	**0.176**
Fish	4	0.295 (3)	0.269	0.198	**0.161**
Frog	5	0.289 (4)	0.271	0.183	**0.169**
Horse	5	0.270	0.266	0.198	**0.188**

ACT with two more seeds gives 0.201 and 0.218 mm on the same 48 drawings (three seeds: 0.204 ± 0.013 mm), all well below both IK variants.

Is it the learning or the settle gate? Streaming IK uses ACT's progress rule with no settle gate. It finishes all 48 drawings (IK's timeouts came from the gate), but its path RMSE improves on IK by only 0.010 mm (0.267 mm, lower on 37 of 45), its corner error grows from 0.366 to 0.608 mm because pure pursuit cuts corners, and its jerk stays at IK's level. ACT is lower than streaming IK on all 48 drawings (paired reduction 0.075 mm, CI 0.069 to 0.080). Slowing streaming IK to IK's own pace (carrot limit 6.9 mm s^{-1}) does not help: 0.266 mm RMSE, jerk down to 524 but corner error up to 0.973 mm, and 181 s per drawing, still above ACT on all 48 drawings. Removing the gate is not what makes ACT

TABLE IV

HOW THE ACT PATH FOLLOWER GOT THERE (48 GOLD DRAWINGS, 1 G, MEANS OVER FINISHED DRAWINGS, RUNS MADE DURING DEVELOPMENT WITH EARLIER CODE). ROWS ABOVE THE RULE USE THE DEFAULT SIM SCENE, ROWS BELOW THE STIFFENED LAB-ARM TWIN.

Controller	Fin.	RMSE	Max	Jerk
Lula IK	39/48	0.698	0.985	806
ACT, 30 Hz, full step held 2 ticks	48/48	0.366	2.599	1018
+ 60 Hz queries, half step (Eq. 3)	48/48	0.325	2.360	272
+ progress capped at 5 mm (hairpins)	48/48	**0.239**	1.472	**255**
+ cap at 1.5 mm (spike tips)	48/48	0.256	**0.885**	285
Lula IK, lab-arm twin	44/48	0.283	**0.775**	752
ACT, lab-arm twin, height jitter	**48/48**	**0.195**	0.812	**390**

TABLE V

PAPER PLACEMENT ROBUSTNESS, 16 GOLD DRAWINGS (FINISHED, MEAN PATH RMSE IN MM).

Placement	Lula IK		ACT	
Nominal	15/16	0.278	16/16	0.192
12 mm lower	13/16	0.284	16/16	0.189
12 mm higher	15/16	0.271	16/16	0.193
Shifted $(+15, -15)$ mm	16/16	0.300	16/16	0.190

accurate; the learned policy is.

What mattered. Table IV traces the design. Querying at 60 Hz with half steps cut jerk 3.7×; capping the progress projection fixed hairpin jumps (mean worst point 2.36 to 1.47 mm), and a tighter cap fixed spike tips (0.885 mm) at a small RMSE cost. Moving to the stiffened lab-arm twin, with paper-height jitter in training, gives the final policy. Each variant beats IK on path RMSE in its own scene.

Paper placement. Table V moves the sheet on a 16-drawing subset. ACT, trained with height jitter, finishes all 64 runs within 0.189 to 0.193 mm. IK drops to 13/16 with the paper 12 mm lower and degrades to 0.300 mm under a planar shift.

SmolVLA against ACT. On the full gold set with the batch runner (Table II), SmolVLA stage 2, served with zero noise, three flow steps and a CUDA graph and re-planned every tick, is slightly more accurate than ACT: 0.180 against 0.192 mm, lower on 36 of 48 drawings (paired reduction 0.012 mm, CI 0.008 to 0.017, $p = 5 \times 10^{-6}$), and best in every class. It pays for it with a third more joint jerk (521 against 390), a higher worst point (0.871 against 0.776 mm), and twice the query latency (17.7 against 9.1 ms). In the interactive runner used for the studies below the order was reversed (0.230 against 0.206 mm on six animals, 10 flow steps, and 0.252 against 0.206 mm on one cat with the same three-step CUDA-graph server). The runners share the policy servers but have separate control loops, and we have not isolated which difference flips the order, so we compare only within a runner. The interactive-runner studies use one gold drawing per class and six face sketches unseen by both policies (Table VI). Offline, on 1,024 held-out frames, ACT's first-action error is 0.28 (normalised so that predicting zero scores 1.0) against 0.36 for SmolVLA. In closed loop (10 flow steps), executing three actions per chunk gives 0.327 mm; re-planning every tick with the same checkpoint gives 0.230 mm, within 12% of ACT. Most of the gap was therefore execution, not prediction. On faces, SmolVLA is 14% behind ACT. Both generalise from animals to faces at about +22 to +25% RMSE. Faces were never in either policy's training data.

TABLE VI

SMOLVLA AND ACT, INTERACTIVE RUNNER. MEAN PATH RMSE / P95 (MM); ALL RUNS FINISHED.

Controller	6 animals	6 faces
SmolVLA, 3 actions per chunk	0.327 / 0.581	-
SmolVLA, re-plan every tick	0.230 / 0.410	0.288 / 0.473
ACT	**0.206 / 0.346**	**0.252 / 0.411**
Hybrid ACT, no skips	0.210 / 0.336	0.253 / 0.410
Hybrid ACT, injected skips	0.212 / 0.344	0.256 / 0.421
Hybrid SmolVLA, injected skips	0.244 / 0.409	-

End-to-end use. The pipeline and controllers are wrapped in WildTrace Studio, a web application in which a user asks for an animal in chat or uploads a photo, picks a face, and watches the twin draw it live with camera feeds and a live pen trace. On a 21-stroke portrait started this way, ACT drew at 0.241 mm in 143 s of simulated time against IK's 0.408 mm in 286 s; reaching nearby strokes at pen-up height rather than full approach height then cut ACT's time by 27% with unchanged error.

Camera-conditioned SmolVLA (negative result). Replacing the rendered path with synthetic overhead and tip-crop camera views improved offline error (0.306) but not closed-loop tracking, and the policy ignored the page: blanking it changed first-action error by less than 0.001. Gap-labelled data, two DAgger rounds [22], retrace weighting and an explicit redraw mode taught the policy to reverse direction at a gap (2% of its going-back actions pointed the wrong way, against 38% for ACT), but not to land back on the line: the camera policies abandoned 16 of 17 self-directed redraws.

Hybrid repair. With the pen made to skip ink on purpose (3 to 8 mm skips every 60 mm of inked travel on average), the supervisor made 87 redraws for 105 skips under ACT across 12 drawings, gave up none, and left at most 0.49% of any path uninked; the remaining skips fell on line already re-inked or were shorter than the gap threshold. The cost was at most 0.011 mm RMSE and 0.4 to 2.2 s per drawing. Without skips the supervisor never intervened.

Speed. Scaling ACT's applied step (Fig. 7) from 0.5 to 0.75 of the prediction keeps RMSE within 6% (0.206 to 0.219 mm) and cuts simulated time by 24%; 1.0 gives +11% error for 38% less time; 2.0 overshoots and trips the 5 mm safety stop, consistent with the curvature-speed trade-off reported for Piktor-O-Bot [13]. SmolVLA has no faster setting within 10%.

Physical arm. The twin-trained ACT policy ran on the lab arm without fine-tuning. In a 30-drawing batch (five per class, not the gold-set drawings) every drawing finished, at 0.532 mm mean path RMSE (0.421 to 0.622 mm), 2.7× the twin (Table VII, Fig. 8). Stalls were the main failure mode: 14 of 30 drawings needed at least one nudge to resume (26 in

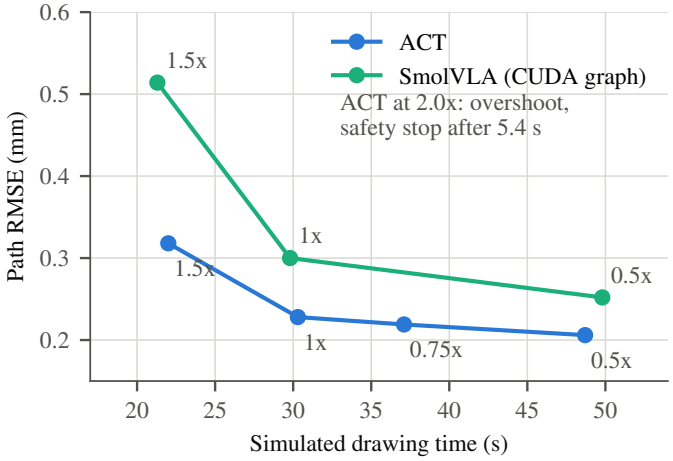


Fig. 7. Speed against accuracy on the gold cat (interactive runner): the step scale α of Eq. (3) relative to the default 0.5 is shown next to each point. ACT keeps most of its accuracy up to 1.0; SmolVLA degrades sooner.

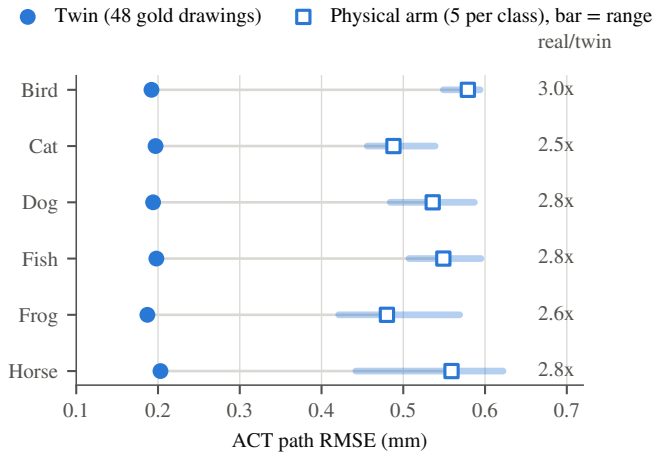


Fig. 8. ACT path RMSE per class in the twin (gold set) and on the physical arm (five drawings per class, range shown). The real arm is 2.5 to 3.0× the twin in every class. Drawings differ between the two sets, so the comparison is per class, not paired.

TABLE VII
ACT ON THE PHYSICAL ARM, 30 ANIMAL DRAWINGS (FIVE PER CLASS). PATH RMSE IN MM.

Class	RMSE mean (range)	Worst point	Time (s)	Nudges / drawing
Bird	0.579 (0.549-0.594)	1.54	76	0.8
Cat	0.488 (0.456-0.539)	1.53	96	1.6
Dog	0.536 (0.484-0.587)	1.47	72	0.0
Fish	0.549 (0.507-0.595)	1.49	102	0.4
Frog	0.480 (0.421-0.569)	1.51	100	0.4
Horse	0.559 (0.442-0.622)	1.54	117	2.0
All	0.532 (0.421-0.622)	1.54	94	0.87

TABLE VIII
FACE PORTRAITS ON THE PHYSICAL ARM WITH ACT AND A MARKER. RMSE, P95 AND MAX ARE MEASURED FROM THE ARM'S REPORTED TIP AT 10 HZ; CHAMFER AND COVERAGE ARE MEASURED FROM THE INK IN AN OVERHEAD PHOTOGRAPH. MM.

Run	RMSE	p95	Max	Chamfer	Cov. 2 mm	min
A (48 strokes), used sheet	0.486	0.912	1.306	0.74	90%	8.5
A (48 strokes), clean sheet	0.514	0.966	1.305	0.67	90%	8.3
B (59 strokes), clean sheet	0.490	0.905	1.273	2.13	71%	9.7

total), always mid-stroke where waypoints are dense; the pen sat about 0.5 mm beside the path and corrected only sideways. Three face portraits drawn with markers (Table VIII) finished at 0.497 mm measured RMSE, using an automatic 0.5 s stall release instead of nudges.

Scoring the ink. The arm's reported tip says where the arm believes the pen is, not what reached the paper. We therefore score the sheet itself: the page is found in an overhead photograph and rectified to millimetres, the ink is thinned to a centre line, and the ink is fitted to the plan by a trimmed rigid transform (ink onto plan, so missing ink cannot drag the fit). On portrait A the ink lies 0.43 to 0.48 mm from the plan on average, in line with the reported tip error, and 90% of the plan has ink within 2 mm. Portrait B scores lower for a reason the arm's log cannot see: the long hair below the shoulders inked faintly, so only 71% of the plan is covered. The photograph's 0.6 mm pixels bound this measurement; flatbed scans are planned for the paired real-arm study.

VI. DISCUSSION AND LIMITATIONS

Why ACT beats its own teacher. ACT is trained on IK labels yet outperforms IK because the labels are kinematic: they describe where the arm should go, not IK's gravity-induced sag, and the noise-injected episodes teach correction. Training on physics rollouts of IK imported IK's error. Querying a short path window at 60 Hz also replaces IK's stop-and-settle with continuous motion, which is where the jerk and the tremor come from. The streaming IK result shows that continuous motion alone is not enough: an IK solver chasing a moving target cuts corners and stays jerky, while the policy has learned from noise-injected labels how to correct toward the path.

Sim-to-real gap. The 2.7× gap is consistent across classes, which points to a systematic cause rather than content. Candidates we observed are a non-level sheet (line weight varies across the page), pen compliance, the arm's own position controller differing from the twin's servo, and dense-waypoint stalls. Fitting the paper plane from a three-point touch and training on stalled states are the next steps.

What the pipeline does not preserve. FLUX passes the referee reliably but redraws the subject in a generic pose: in our generator study its median silhouette IoU with the source photo was about 0.2, against 0.78 to 0.91 for OmniGen2's non-blank drawings, and facing direction matched in 65% of cases. The drawings are recognisably the right animal, not a tracing of the photo. Lower diffusion strength or contour conditioning is the obvious next step.

Baseline strength. We compare against the waypoint-gated IK controller the system shipped with and a streaming IK that shares ACT's progress rule; an optimisation-based tracker (for example model-predictive contouring control) and the vendor's own Cartesian motion commands on the real arm are stronger

baselines we have not yet run.

Limitations. Each controller was run once per drawing; ACT was trained with three seeds; the other policies with one. The real-arm batch used different drawings from the gold set and relied on nudges, so it is not a controlled paired comparison. The path metric is geometric; ink scoring from photographs covers only three portraits so far, and perceived quality (a user study) is future work. The IK baseline is tolerance-gated and its timeouts could be reduced by tuning. Real-arm portraits take 8 to 10 min, against 43 s to 2 min for systems built for exhibition speed [9]–[11]; our runs used the default step scale and stall releases, not a speed-tuned setting. The hybrid supervisor and the camera policies were evaluated on 12 drawings in the interactive runner only.

Reproducibility. Every number in this paper comes from per-run metric files kept with the code: the configuration hash, checkpoint and runner of each run are recorded, and one script regenerates all figures and the statistics in Tables II and III.

VII. Conclusion

WildTrace turns photographs into robot pen drawings end to end. A compact ACT path follower trained only on kinematic IK labels draws held-out content more accurately, more reliably and more smoothly than the IK solver that labelled it, is robust to paper placement, and transfers to a physical xArm 7 at about half a millimetre RMSE. A streaming IK with the same progress rule gains almost nothing, so the accuracy comes from learning. A fine-tuned VLA re-planned every tick is slightly more accurate again but jerkier and slower to query, and a simple supervisor makes either policy robust to missed ink. Closing the remaining sim-to-real gap is our next goal.

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